Assembly Language for x86 Processors

7th Edition , Global Edition

Kip R. Irvine

Chapter 15: Disk Fundamentals

(c) Pearson Education, 2015. All rights reserved. You may modify and copy this slide show for your personal use, or for use in the classroom, as long as this copyright statement, the author's name, and the title are not changed.

Chapter Overview

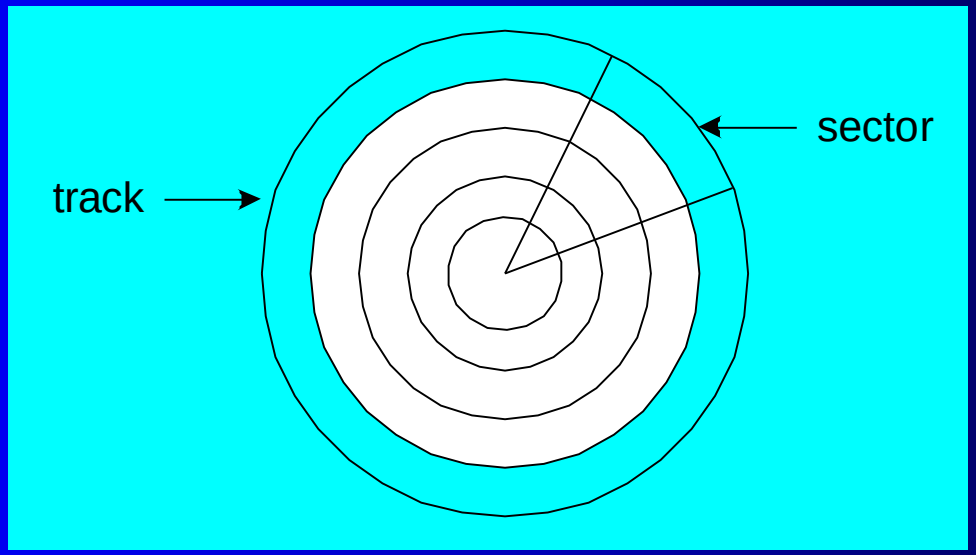
- **Disk Storage Systems**
- File Systems
- Disk Directory
- Reading and Writing Disk Sectors (7305h)
- System-Level File Functions

Disk Storage Systems

- Tracks, Cylinders, and Sectors

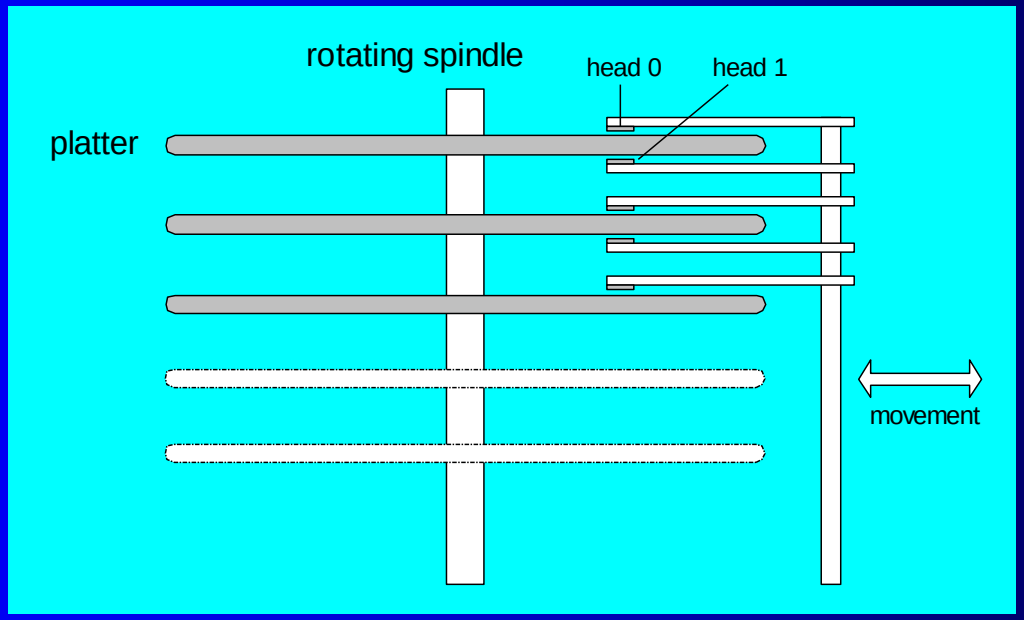
Tracks and Sectors

- Physical disk geometry - a way of describing the disk's structure to make it readable by the system BIOS
- Track - concentric circle containing data
- Sector - part of a track



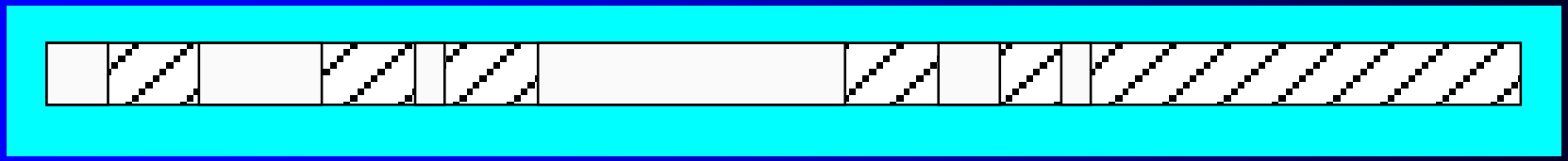
Cylinders and Seeking

- Cylinder - all tracks readable from one head position
- Seek - move read/write heads between tracks



Fragmentation

- A fragmented file is one whose sectors are no longer located in contiguous areas of the disk.
  - causes read/write heads to skip
  - slower file access
  - possible read/write errors



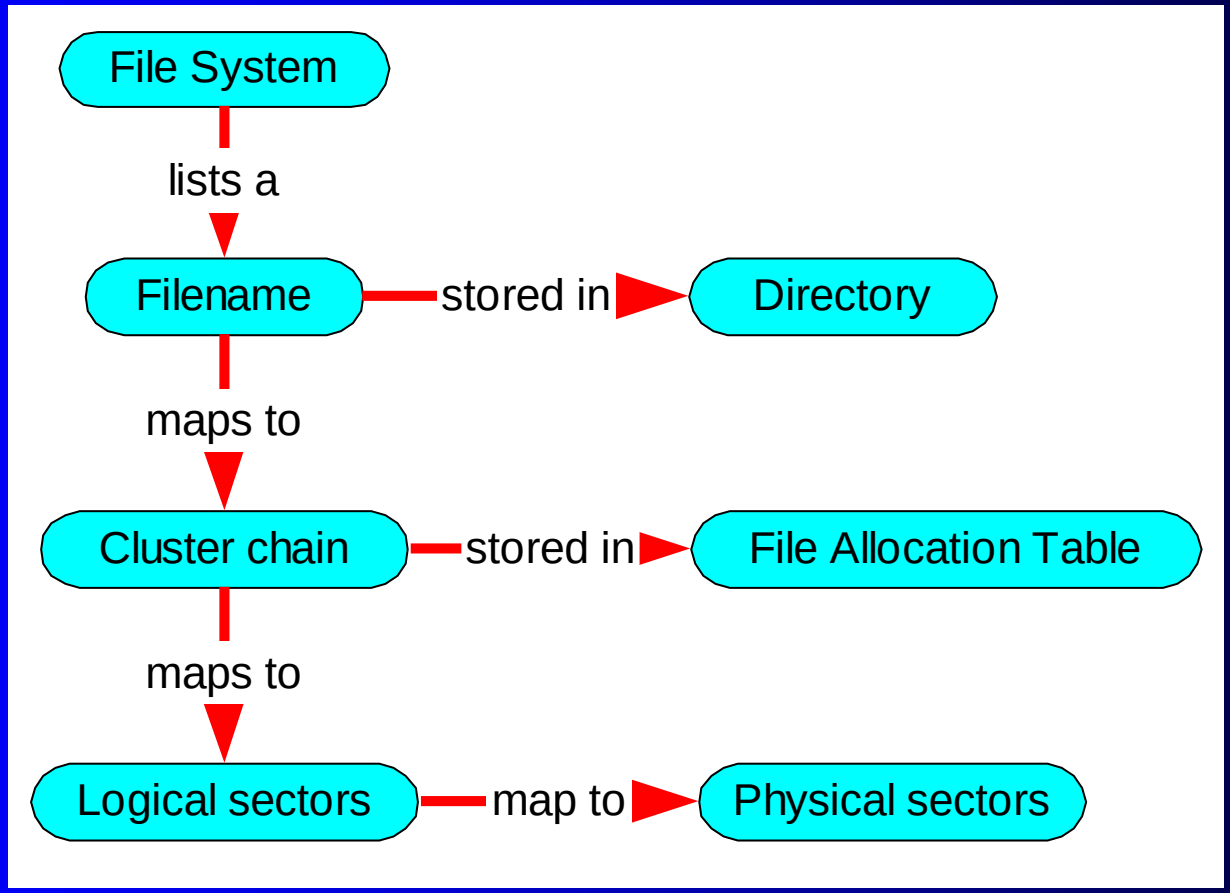
What's Next

- Disk Storage Systems
- **File Systems**
- Disk Directory
- Reading and Writing Disk Sectors (7305h)
- System-Level File Functions

File System

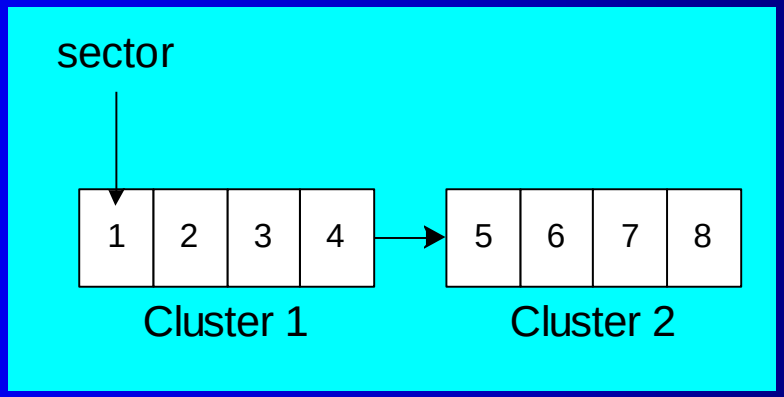
- This is what it does for you:
  - Keeps track of allocated and free space
  - Maintains directories and filenames
  - Tracks the sector location of each file and directory

Directory, File, Cluster, Sector Relationships



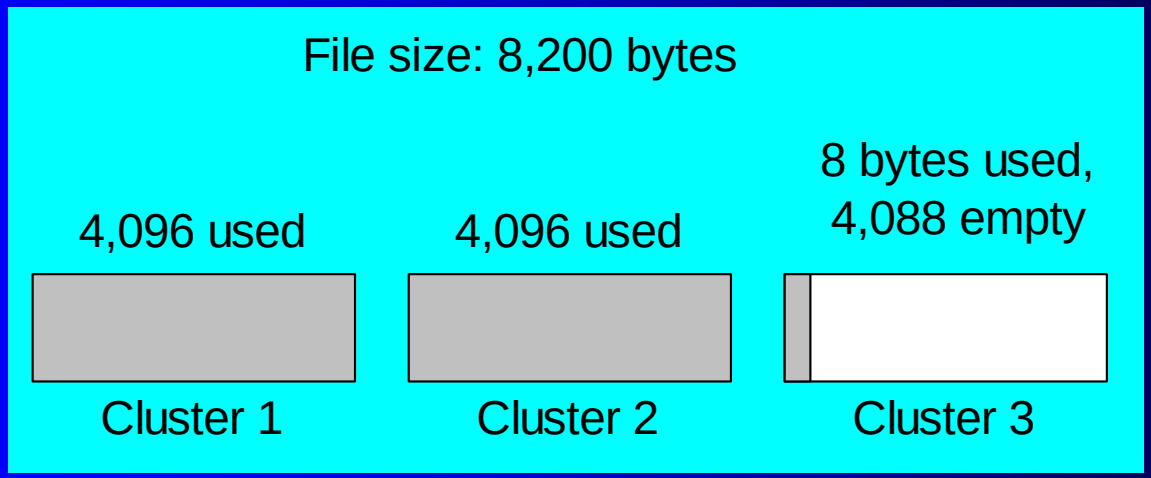
Cluster (1 of 2)

- Smallest unit of space used by a file
- Consists of one or more adjacent sectors
- Size depends on both the type of file system in use and the disk partition size
- A file is a linked sequence of clusters. Example:



Cluster (2 of 2)

- A file always uses at least one cluster
- A high percentage of space may be wasted
- Example: 8,200-byte file requires three 4K clusters:



What's Next

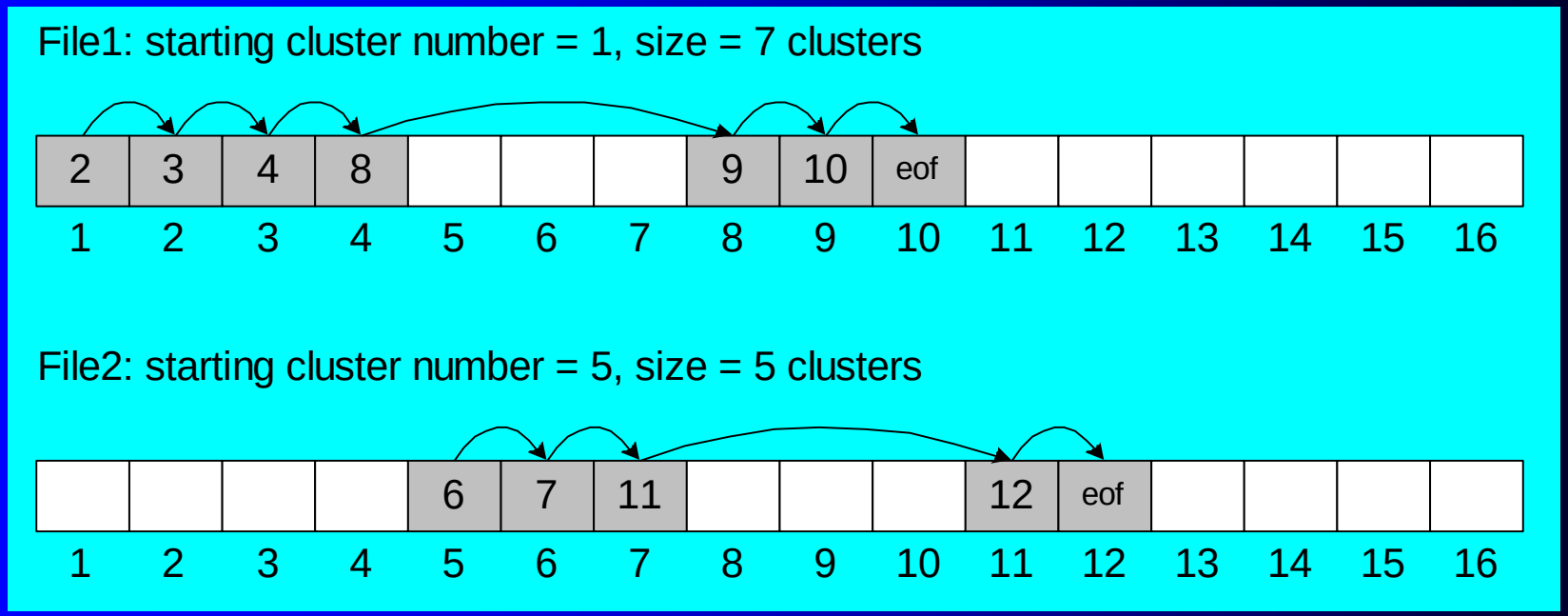
- Disk Storage Systems
- File Systems
- **Disk Directory**
- Reading and Writing Disk Sectors (7305h)
- System-Level File Functions

File Allocation Table (1 of 2)

- A map of all clusters on the disk, showing their ownership by specific files
- Each entry corresponds to a cluster number
- Each cluster contains one or more sectors
- Each file is represented in the FAT as a linked list, called a cluster chain.
- Three types of FAT's, named after the length of each FAT entry:
  - FAT-12
  - FAT-16
  - FAT-32

File Allocation Table (2 of 2)

- Each entry contains an *n*-bit integer that identifies the next entry. (*n*=12,16, or 32)
- Two cluster chains are shown in the following diagram, one for File1, and another for File2:



What's Next

- Disk Storage Systems
- File Systems
- Disk Directory
- **Reading and Writing Disk Sectors (7305h)**
- System-Level File Functions

Reading and Writing Disk Sectors (7305h)

- INT 21h, Function 7305h (absolute disk read and write)
- Reads and writes logical disk sectors
- Runs only in 16-bit Real-address mode
- Does not work under Windows 2000, XP, Vista, Windows 7
  - Tight security!

DISKIO Structure

- Used by Function 7305h:

```
DISKIO STRUCT
   startSector DWORD 0       ; starting sector number
   numSectors  WORD 1        ; number of sectors
   bufferOfs   WORD buffer   ; buffer offset
   bufferSeg   WORD @DATA    ; buffer segment
DISKIO ENDS
```

Example

Example: Read one or more sectors from drive C:

```
.data
buffer BYTE 512 DUP(?)
diskStruct DISKIO <>
.code
    mov ax,7305h              ; absolute Read/Write
    mov cx,0FFFFh             ; always this value
    mov dl,3                  ; drive C
    mov bx,OFFSET diskStruct
    mov si,0                  ; read sector
    int 21h
```

What's Next

- Disk Storage Systems
- File Systems
- Disk Directory
- Reading and Writing Disk Sectors (7305h)
- **System-Level File Functions**

System-Level File Functions

- Common Disk-Related Functions
- Get Disk Free Space
- Create Subdirectory
- Remove Subdirecrory
- Set Current Directory
- Get Current Directory

Common Disk-Related Functions

Function Number	Function Name
0Eh	Set default drive
19h	Get default drive
7303h	Get disk free space
39h	Create subdirectory
3Ah	Remove subdirectory
3Bh	Set current directory
41h	Delete file
43h	Get/set file attribute
47h	Get current directory path
4Eh	Find first matching file
4Fh	Find next matching file
56h	Rename file
57h	Get/set file date and time
59h	Get extended error information

Create Subdirectory

```
.data
pathname BYTE "\ASM",0

.code
    mov ah,39h                      ; create subdirectory
    mov dx,OFFSET pathname
    int 21h
    jc  DisplayError
    .
    .
DisplayError:
```

Remove Subdirecrory

```
.data
pathname  BYTE  'C:\ASM',0

.code
    mov ah,3Ah                    ; remove subdirectory
    mov dx,OFFSET pathname
    int 21h
    jc  DisplayError
    .
    .
DisplayError:
```

The End

